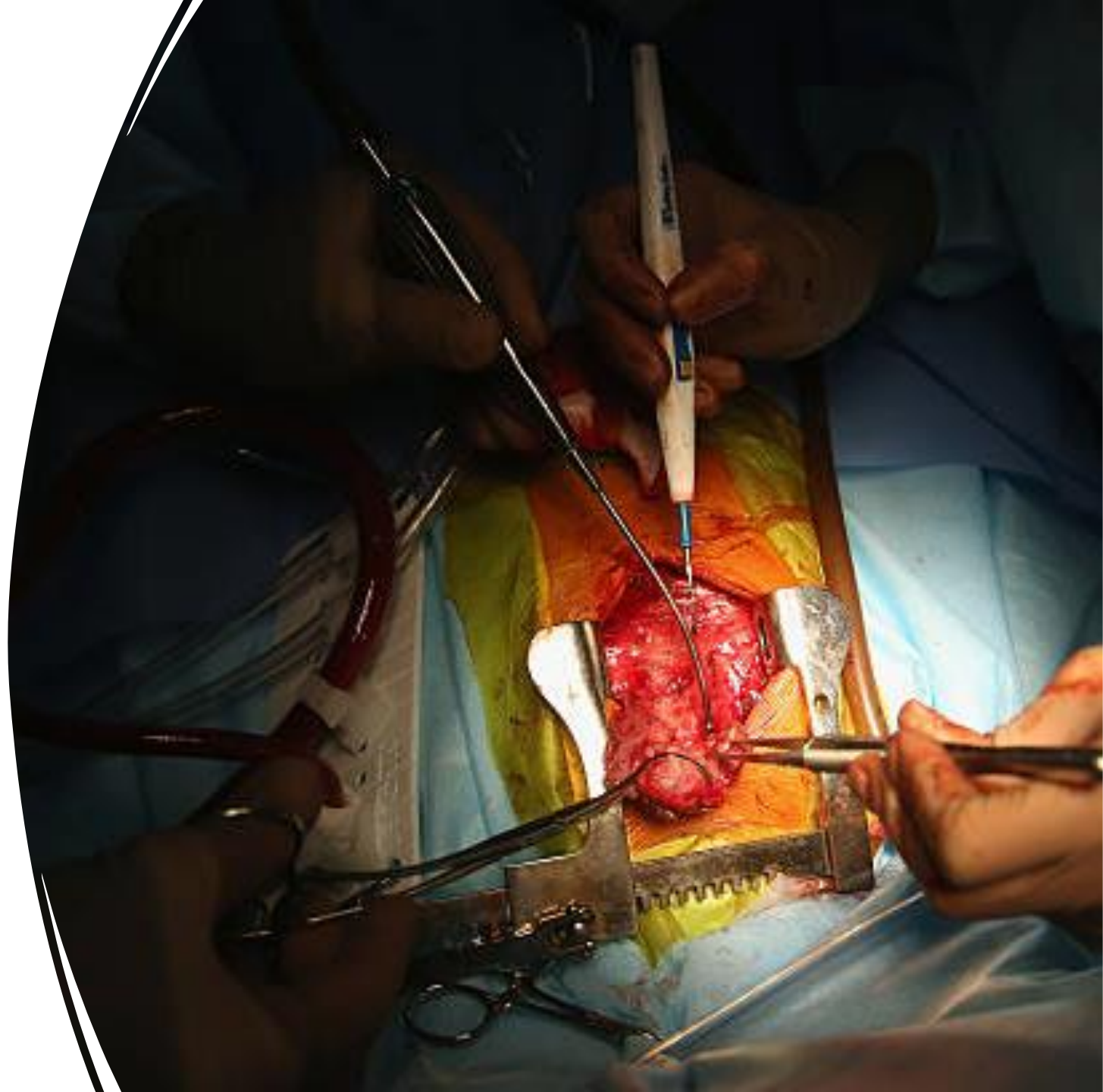


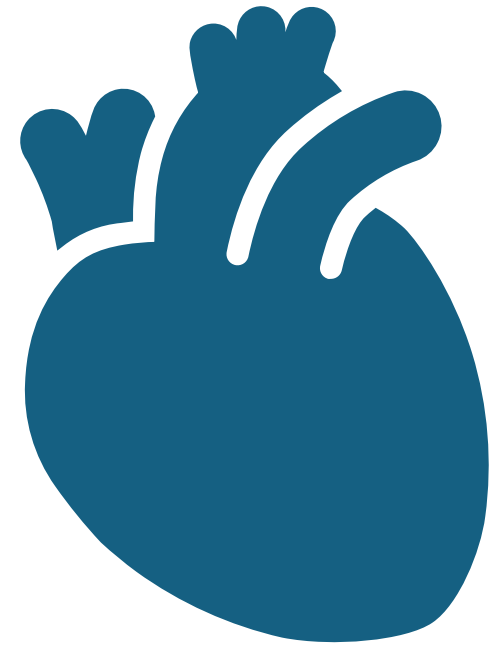
Cardiac surgery

Prepared by
Dr Ghada Elsayed



Outlines:

- Anatomy of the heart:
- Definition of Open Heart Surgery
- Indication of open heart surgery
- Recent techniques in open heart surgery
- What is a heart-lung machine?
- What is Off-pump surgery?
- What are cooling techniques?
- Open heart Surgery
Complications
- Nursing Care of the Post-
Operative for open heart surgery
Patient



Pre test

Mention
indication of
cardiac surgery

Mention
complication
post cardiac
surgery

Definition of Open Heart Surgery:

Open heart surgery is a type of heart surgery that requires the heart to be opened in order to perform the cardiac intervention.

In these cases, the sternum (breast bone) is cut in half vertically so the chest can be opened, making the heart visible.

The pericardium is then pulled away from the heart, giving the surgeon direct access.

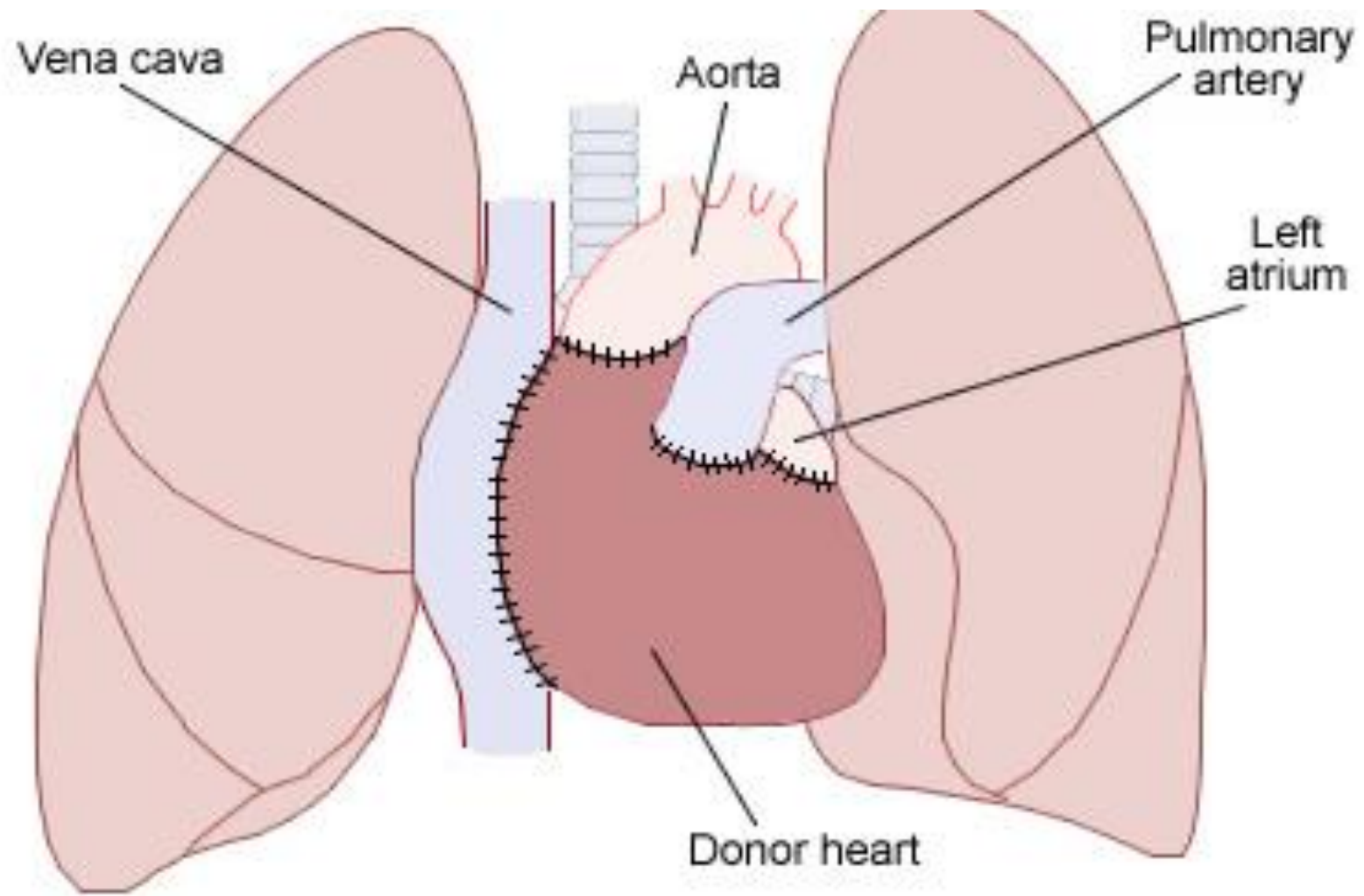
Indication of open heart surgery:

Treat complications of ischemic heart disease (for example, coronary artery bypass grafting),

Correct congenital heart disease, A congenital heart defect (CHD) is a defect in the structure of the heart and great vessels which is present at birth.

Valvular heart disease (the aortic and mitral valves on the left and the pulmonary and tricuspid valves on the right).

Heart transplantation. Hearts transplants or cardiac transplantation is a surgical transplant procedure performed on patients with end-stage heart failure or severe coronary artery disease.



Recent techniques in open heart surgery:

- Newer techniques allow heart surgery to be done through much smaller surgical cuts than the large cut needed for open surgery.
- The heart surgeon will make a 2-inch to 5-inch-long surgical cut in the chest wall.
- Muscles in the area will be divided so the surgeon can reach the heart.
- The surgeon can repair or replace a valve or perform bypass surgery.
- During endoscopic surgery, the surgeon makes one to four small holes in the chest.
- Then the surgeon uses special instruments and a camera to perform the surgery.

What is a heart-lung machine?

The heart-lung machine is also called a cardiopulmonary bypass machine.

It takes over for the heart by replacing the heart's 'pumping action and by adding oxygen to the blood.

This means that the heart will be still for the operation, which is necessary when the heart has to be opened (open heart surgery).

Because the heart-lung machine takes over the work of the heart, surgeons can operate on a heart that is not moving or full of blood.



When the patient is connected to the heart-lung machine, it does the same job that the heart and lungs would do.



The heart-lung machine carries blood from the upper-right chamber of the heart (the right atrium) to a special reservoir called an oxygenator.



Inside the oxygenator, oxygen bubbles up through the blood and enters the red blood cells.



This causes the blood to turn from dark (oxygen-poor) to bright red (oxygen-rich).



Then, a filter removes the air bubbles from the oxygen-rich blood, and the blood travels through a plastic tube to the body's main blood conduit (the aorta).



From the aorta, the blood moves throughout the rest of the body.



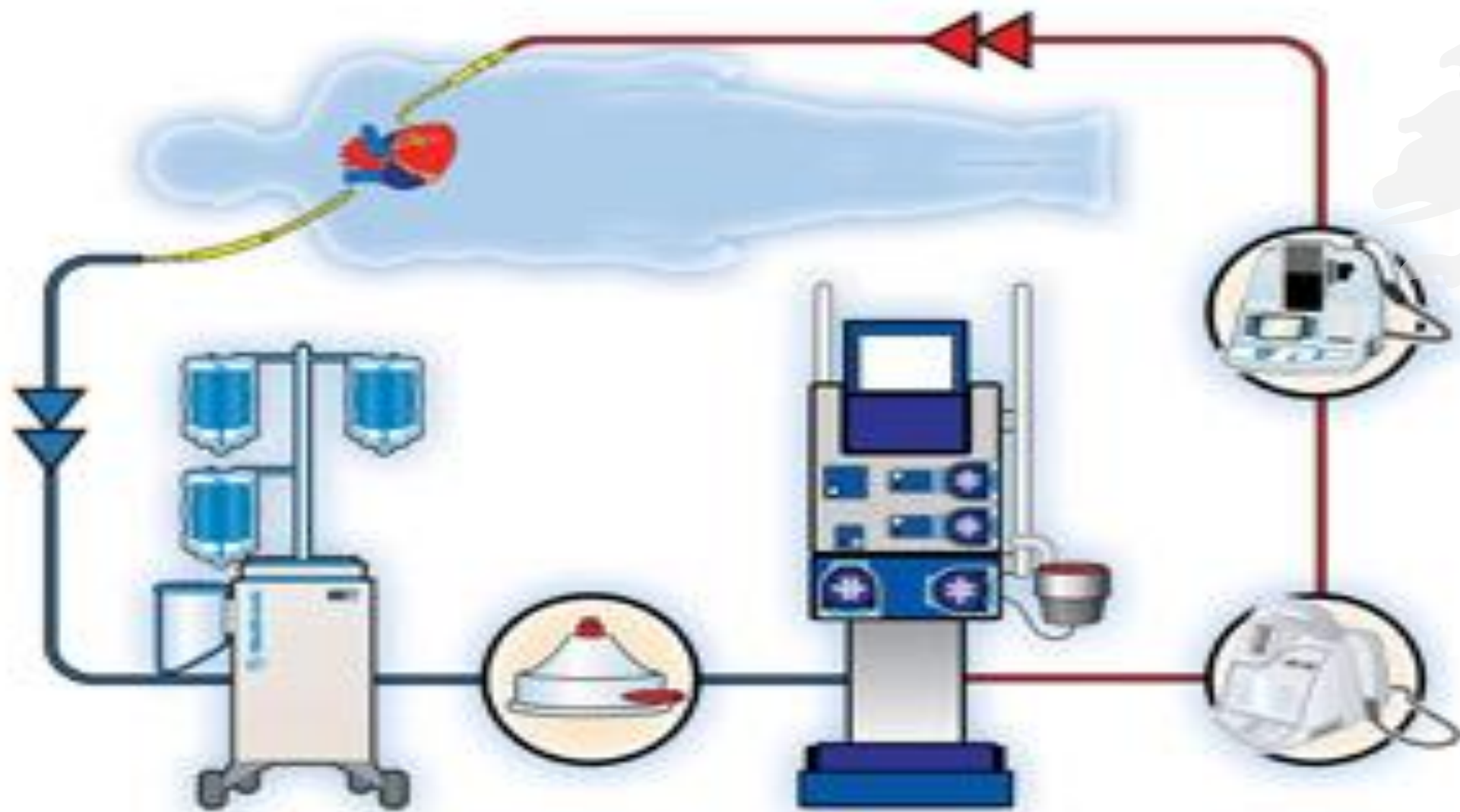
The heart-lung machine can take over the work of the heart and lungs for hours.



Trained technicians called perfusion technologists (blood flow specialists) make sure that the heart-lung machine does its job properly during the surgery.



Even so, surgeons still try to limit the time that patients must spend hooked up to the machine.





What is Off-pump surgery?

- Off-pump coronary artery bypass or "beating heart" surgery is a form of [coronary artery bypass graft](#) (CABG) surgery performed without [cardiopulmonary bypass](#) (heart-lung machine) as a treatment for [coronary heart disease](#).
- During most bypass surgeries, the heart is stopped and a heart-lung machine takes over the work of the heart and lungs.
- When a cardiac surgeon chooses to perform the CABG procedure off-pump, also known as OPCAB (Off-pump Coronary Artery Bypass), the heart is still beating while the graft attachments are made to bypass a blockage

Off-pump coronary artery bypass was developed partly to avoid the complications of cardiopulmonary bypass during cardiac surgery.

The potential benefits/advantages of off-pump surgery may include the following:

Reduced need for blood transfusions

Reduced risk of bleeding, stroke and kidney failure

Potential for reduced psychomotor and cognitive problems

High-risk patients with additional diseases like lung disease, kidney failure and peripheral vascular disease may benefit from this kind of operation

What are cooling techniques?

Cooling techniques let surgeons stop the heart for long periods without damaging the heart tissue.

Cool temperatures avoid damage to the heart tissue by reducing the heart's need for oxygen.

The heart may be cooled in 2 ways:

Blood is cooled as it passes through the heart-lung machine. In turn, this cooled blood lowers body temperature when it reaches all of the body parts.

Cold salt-water (saline) is poured over the heart.



After cooling, the heart slows and stops.



Injecting a special potassium solution into the heart can speed up this process and stop the heart completely.



The heart is then usually safe from tissue injury for 2 to 4 hours.



Complications of open-heart Surgery:



Complications During and After Heart Surgery

Bleeding: May occur at the incision site or from the area of the heart where surgery is performed

Abnormal heart rhythm: In rare cases a temporary external, or permanent internal pacemaker may be necessary to correct this problem.

Ischemic heart damage: Damage to heart tissue caused by a lack of blood flow to the heart

Death: The risk of death is increased in surgeries where the heart is stopped for the procedure.

Complication of cardiac surgery

- **Blood clots**: Clots may form in and around the heart or travel through the bloodstream.
- **Stroke**: Often caused by clots that form in the blood after surgery; Impairment of the brain function may be caused as result of blood clots, which in turn restricts the blood and oxygen supply to the brain.
- **Cardiac tamponade (pericardial tamponade)**: A life-threatening condition where the pericardium, the sac surrounding the heart, fills with blood.
- This makes it difficult, or impossible, for the heart to fully function.

Lung Complications:



As all known, the functioning of lungs and heart are related indirectly.



In open heart surgery, patients lung problems do develop after the procedure, respiratory tract infections, bleeding and in severe cases, lung failure.



Minor Problems

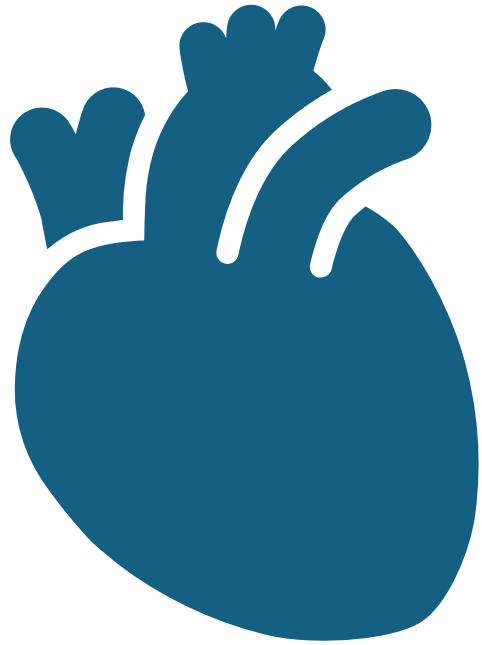
Heart surgeries may result in temporary problems like



Nausea, vomiting,



Bruising, inflammation, scar formation, skin numbness, skin allergic reactions



Global Overview about Before, During and After Open Heart Surgery:

- 1) Preoperative instruction and preparation for Open Heart Surgery:
- **Before having open heart surgery,**
- The patient will meet with the cardiothoracic surgeon, who will already have reviewed the tests
- (such as films from the angiography) and will explain the results of these tests, what the surgery entails, the benefits the patient can expect from surgery, and, especially, the risks of surgery.
- This meeting provides the patient a chance to voice fears, ask questions, and perhaps look over the angiogram film to develop a better understanding of why the operation is necessary.



Approximately two weeks before the surgery,

- The patient may be instructed to stop taking any medications that might affect the ability of the blood to clot, including **aspirin, and warfarin (Coumadin)**.
- The patient should continue taking beta blockers and calcium antagonists if he or she is on them and can continue use of nitrates to relieve chest pain.

The day before the operation,



The patient will be expected to check into the hospital and again undergo a battery of tests—chest X-ray, ECGs, blood tests, urinalysis even though he or she may recently have had all these tests.



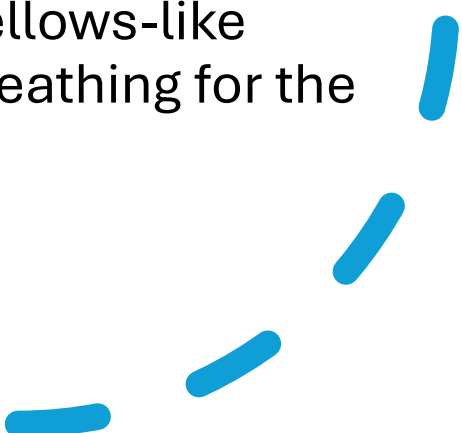
This is an important step to make sure that there has been no change in the status of the patient's health and to ensure that no small but important detail has been omitted.

The night before the surgery

- The patient will be asked **to shower** in order to reduce the bacteria on the skin.
- **Fasting** after midnight, because anesthesia is safer on an empty stomach.
- The **anesthesiologist will usually visit the patient** the night before in order to explain what the anesthesia does and how the anesthesia team takes care of the patient's breathing with the respirator.
- The **chest area will be scrubbed** clean and washed with an antiseptic solution, and body hair will be shaved where necessary.

On the day of surgery:

- **On the morning of surgery,**
- The patient is given a **mild tranquilizer to reduce any anxiety** related to the operation.
- Electrodes **connected to an ECG monitor are then attached to the patient's back** to allow constant monitoring of the heart's electrical activity during the operation.
- Following the administration of a local anesthetic, **intravenous (IV) lines are inserted into the veins of the arm or wrist.**

- 
- A **Foley catheter** is inserted into the patient's bladder before the operation.
 - **Anesthetic agents** are then administered directly into the veins.
 - These agents have three functions: to block pain and induce drowsiness, to relax the muscles and prevent the person from moving and jerking during the operation,
 - Once the patient has been anesthetized, **a tube (endotracheal) is inserted into the patient's** windpipe.
 - The tube connects to the respirator a bellows-like instrument that performs the work of breathing for the patient.
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Another tube (**nasogastric**, or NG) is also inserted to collect stomach fluids that might otherwise nauseate the anesthetized patient.

An **anticoagulant, heparin**, is also administered at the start of the operation.

This “blood-thinner” prevents clots, or emboli, from forming and helps to protect the patient from a stroke.

Once the patient has been prepared, the surgeons begin. For open-heart surgery

After the surgery:



The patient is taken to the intensive care unit (ICU), where specialized nursing care is available around the clock and instruments monitor the heart's electrical activity, blood pressure, temperature, and other vital signs.



The patient's family will generally be allowed to visit at this time, although the patient will be confused from the anesthetic and unable to speak because of a tube in the windpipe (endotracheal) that helps the person breathe with a respirator.



The ICU experience can be disorienting, as there is little to differentiate day and night, and the patient will drift in and out of consciousness.

Nursing Care of the Post- Operative for open heart surgery Patient :



Patients will be cared for in the ICU during the immediate Post-operative period.



After extubation and when hemodynamically stable, they will be transferred to Cardiac Step-Down Unit.



DESIRED PATIENT OUTCOMES:



1. Patient will experience minimal/no complications related to CABG surgery.



2. Patient will achieve maximal level of comfort during the post-operative period.

CLINICAL ASSESSMENT AND CARE:

- **A. PATIENT ARRIVAL**
- 1. The primary RN will take report and initial orders from the anesthesiologist in conjunction with ICU house staff.
- 2. The RN assigned to help the primary RN will:
 - a. Connect patient to the monitoring system.
 - b. Connect chest tubes to wall suction.
 - c. Note type and rate of intravenous infusions, isolate volume line.
 - d. Validate respiratory settings on ventilator with respiratory therapy.
 - e. Attach warming apparatus.
 - f. Autotransfusion per MD order.

B. INITIAL **POST-OP** **ASSESSMENT**

- I) To be completed within 15 minutes of arrival to ICU:
- 1. Perform a brief physical assessment and validate the status of the following with the physician/house staff.
- **a. Cardiovascular Status:**
- 1) Obtain full set of vital signs.
- 2) Obtain a cardiac profile and verify waveforms.
- 3) Verify EKG rhythm, verify evidence of pacer capture if appropriate.
- 4) Note amount/quality of chest tube drainage.
- 5) Note quality of peripheral pulses, temperature and color of extremities.
- 6) Demand pacemaker settings.

b. Respiratory Status:

1

1) Check ETT size and position.

2

2) Auscultate bilaterally for breath sounds, note presence of adventitious breath sounds.

3

3) Obtain O sats via probe.

4

4) Obtain ABG as MD order.

5

5) Note peak pressures, expiratory volumes and minute ventilation on current vent settings.

- **c. Neuromuscular Status:**

- 1) Validate LOC.
- 2) Check for pupil reactivity.
- 3) Check for quality and quantity of motor response.
- 4) Identify type and time of last dose of anesthesia, narcotic or paralyzing agent used.

- **d. Genitourinary Status:**

- 1) Measure urine output.

- **a. Integumentary Status:**

- 1) Note presence and characteristics of wound(s):
 - a) chest
 - b) legs/arms - check ace bandages per MD order
- 2) Note presence and characteristics of exit site(s):
 - a) Catheters
 - b) Drains
 - c) Pacer wires
- 3) Assess pressure points for erythema/potential points of break down.

- **b. Gastrointestinal Status:**

- 1) Check bowel sounds.
- 2) Verify placement of N/G tube, if applicable. Attach to wall suction per MD order.

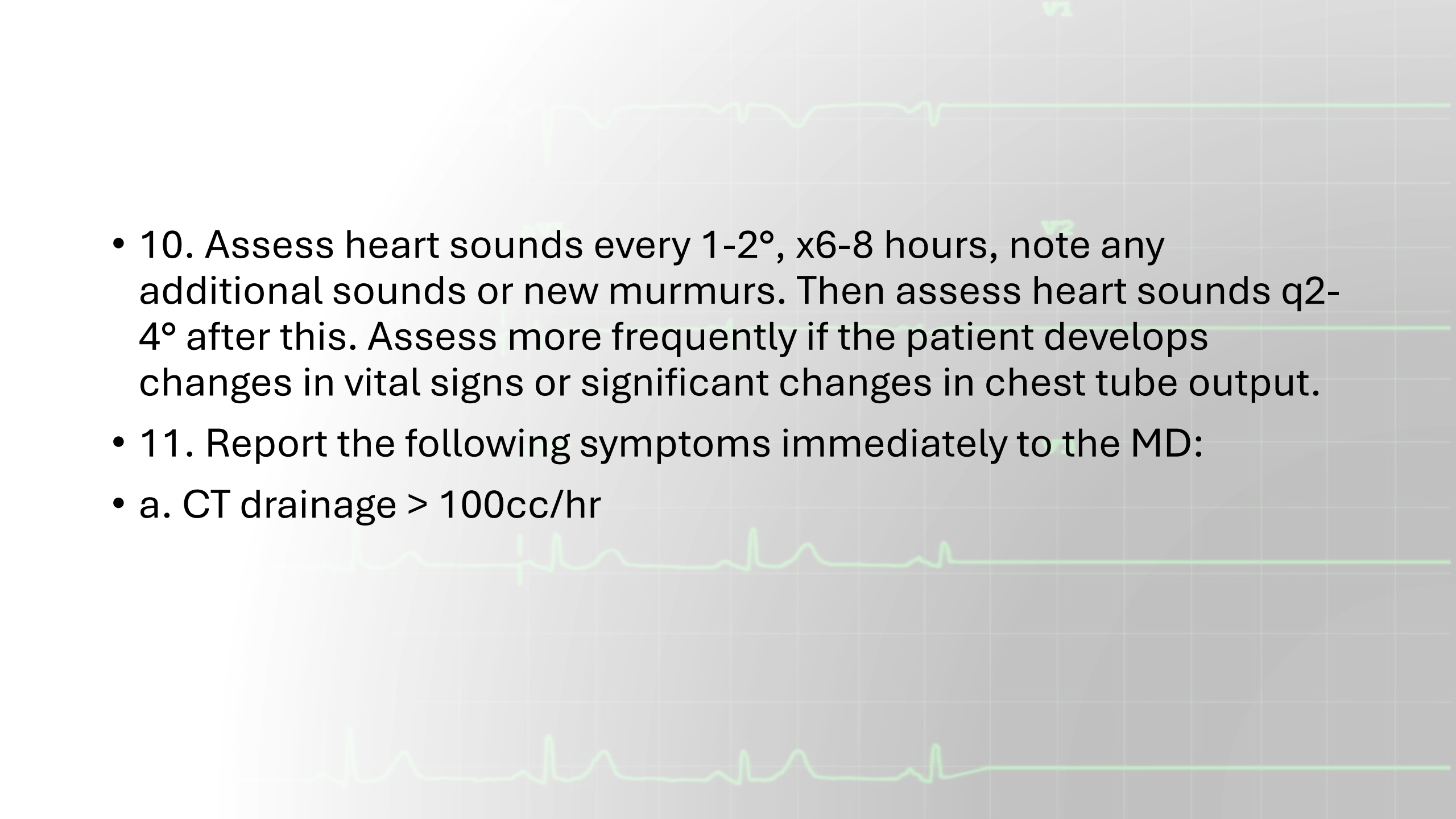
- **C. INITIAL POST-OP CARE:**

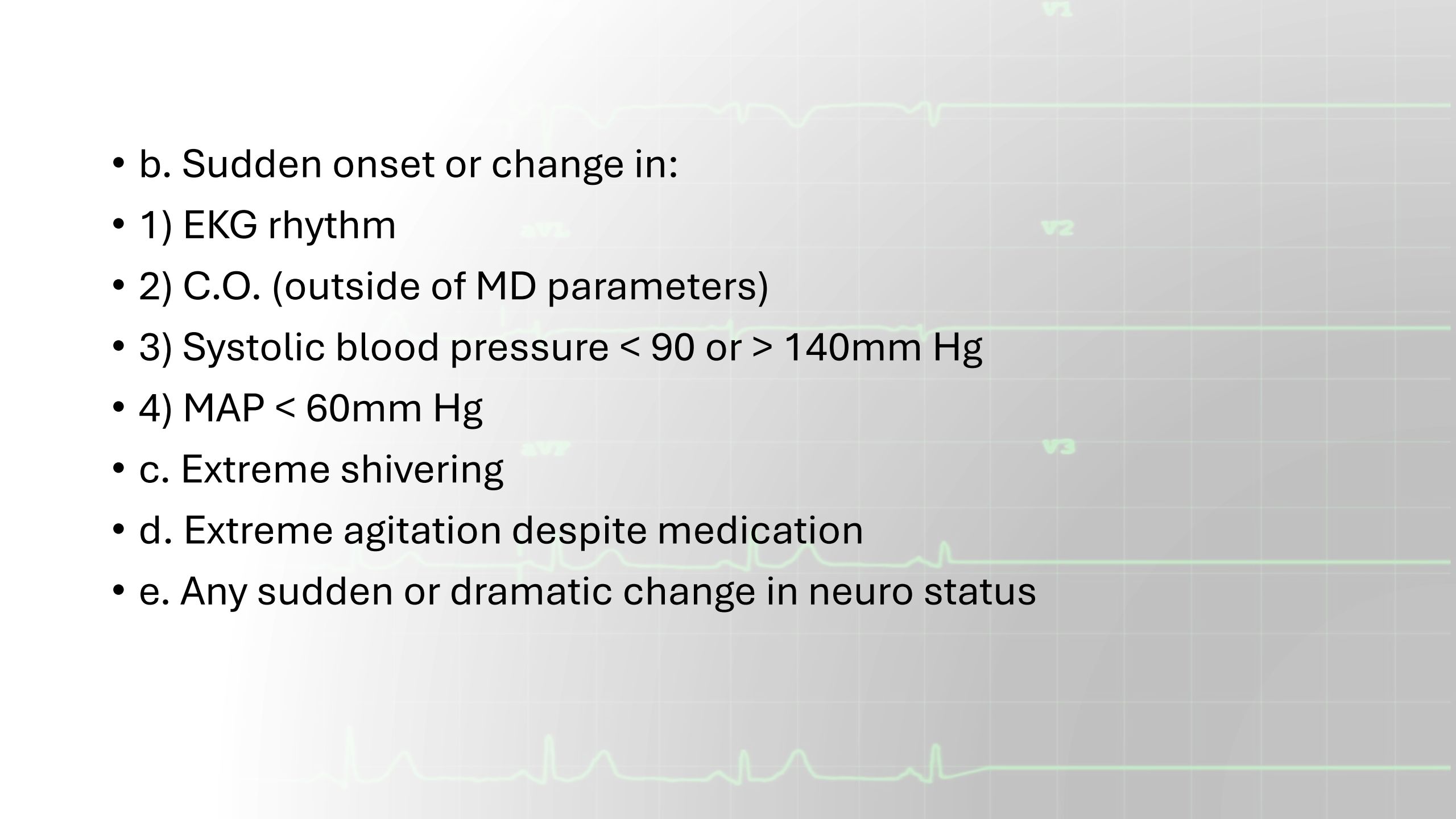
- 1. To be done jointly by primary RN and helper RN within 30 minutes of arrival.
 - 2. Review Post-operative orders, clarify as needed.
 - 3. Obtain 12-lead EKG and chest x-ray as per MD order.
 - 4. Titrate intravenous medication solutions according to MD parameters and based on patient's weight.
 - 5. Initiate measures to warm patient.
 - 6. Obtain cardiac profile, if applicable.
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ONGOING ASSESSMENT

- 1. Obtain vital signs every 15 minutes for 4 hours or prn, every 30 minutes for 4 hours, then every 1 hour until stable.
- a. Obtain core or rectal temperature every 4°.
- b. Discontinue warming measures when temperature reaches 37.5° c.
- 2. Obtain Cardiac profiles as per Hemodynamic monitoring protocol and as per MD order.
- Evaluate need for volume, medications or withdrawal of medications.
- 3. Monitor respiratory status as per Ventilator protocol. Initiate Weaning Protocol when appropriate.
- 4. Monitor temperature of extremities, color, capillary refill, presence of edema, quality of pulses, and motor, function and sensation every 4°.
- 5. Monitor I+O every 1°.
- Assess need for diuretic therapy in collaboration with house staff.

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- 6. Monitor chest tube drainage every 15 minutes for initial 1-2 hours, then q1-2 hour for 24° then every 2-4° until discontinued. From post-op hour 2-24, drainage should decrease to less than 30cc/hour.
 - CT may be discontinued on post-op day 1 when CT output less than 80cc for 8-12°.
 - 7. Monitor neuro status every 1° till stable, then every 2°. Reorient patient to time, surroundings and person as appropriate.
 - 8. Check wounds, pacer wires and dressings every 2°.
 - 9. Assess bowel sounds every 4° while intubated, monitor for flatus.

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- The background of the slide features faint, stylized ECG (heart rate) waveforms in a light green color, overlaid on a light gray grid. The waveforms are positioned horizontally across the top and bottom of the slide, with some peaks and troughs visible. The text is in a black, sans-serif font.
- 10. Assess heart sounds every 1-2°, x6-8 hours, note any additional sounds or new murmurs. Then assess heart sounds q2-4° after this. Assess more frequently if the patient develops changes in vital signs or significant changes in chest tube output.
 - 11. Report the following symptoms immediately to the MD:
 - a. CT drainage > 100cc/hr

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- b. Sudden onset or change in:
 - 1) EKG rhythm
 - 2) C.O. (outside of MD parameters)
 - 3) Systolic blood pressure < 90 or > 140 mm Hg
 - 4) MAP < 60 mm Hg
 - c. Extreme shivering
 - d. Extreme agitation despite medication
 - e. Any sudden or dramatic change in neuro status

g. Poor ABG's or drop in O₂ sat from baseline

h. Dehiscence of wounds

12. Report any of the following promptly to the MD (within 30 minutes):

a. Any evidence of CHF

b. Decreased urine output < 30cc/hr

c. T > 102° F

d. Saturation of chest and/or graft dressing within a short time span (< 1 hour)

e. HCT < 27

E. ONGOING PATIENT CARE

1. Respiratory:

a. Continually monitor SaO sat via probe.

b. Obtain ABG's per MD order.

c. Wean from ventilator as per protocol.

d. Turn, cough and deep breathe every 2°.

e. Post extubation, initiate Respiratory Care Protocol.

f. Incentive spirometer every 1-2° while awake, and PRN post extubation, per MD order.

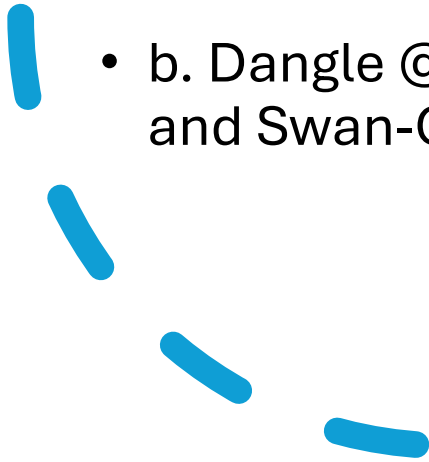
g. Assist patient in use of pillows to splint incision while coughing.

- **2. Cardiovascular:**

- a. Maintain pacer wires insulated and grounded, per MD.
- b. 12-lead EKG and chest x-ray as per MD order.

- **3. Nutrition:**

- a. Remove NG tube post-extubation unless gastroepiploic graft.
- b. Maintain NPO until extubated.
- c. When extubated, progress diet as per MD order.

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- 4. **Neuro:**
 - a. Provide sedation/pain medication as appropriate.
 - 5. **Fluid/Electrolytes:**
 - a. Replace electrolytes as per MD orders.
 - b. Obtain lab tests as per MD orders.
 - c. Weigh patient daily, if ordered by M.D.
 - 6. **Mobility:**
 - a. Turn side-to-side q 2 hours.
 - b. Dangle @ bedside when stable, then out of bed with assist BID when extubated and Swan-Ganz catheter removed.
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PAIN MANAGEMENT:

- 1. Assess patient every 1-2 hours concerning presence of discomfort and characteristics:
 - a. Duration of pain
 - b. Quality (0-10 scale if applicable)
 - c. Location
 - d. Radiation
 - e. Aggravating/alleviating factors
 - f. Quantity
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- 2. Compare and contrast previous angina characteristics versus incisional pain, if applicable.
 - 3. Utilize positioning and support techniques to promote patient comfort, e.g., splinting, pillows, etc.
 - 4. Offer pain med. as needed, per MD order, to achieve patient's identified pain level.
 - 5. Institute alternative methods for distraction as appropriate, i.e., guided imagery.

WOUND CARE:

- 1. Keep dressing clean and dry, and aseptically redress wound. After 24 hours, open to air if there is no drainage.
- 2. Assess for S/S of wound infection:
 - a. Spreading erythema
 - b. Superficial drainage
 - c. Sternal instability (grating or clicking)
 - d. Excessive incisional pain

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- e. Fresh dehiscence
 - g. Purulent drainage
 - h. Positive blood cultures
 - i. Marked leukocytosis
 - j. Signs of systemic sepsis
 - 3. Apply occlusive dressing over CT site when CTs discontinued, x 24°, then D/C.

PATIENT TEACHING:

- 1. Review with patient/family availability of pain meds, how often patient can receive them.
- 2. Instruct patient on cough, deep breathing, exercises, and need for frequent reinforcement.
- 3. Prepare patient/family education.
- 4. Institute cardiac rehab teaching as per protocol; care path/educational booklet.



Cardiac rehabilitation and Health education after open heart surgery:

- Cardiac rehabilitation is a medically supervised program to help heart patients recover quickly and improve their overall physical, mental and social functioning.
 - The goal is to stabilize, slow or even reverse the progression of cardiovascular disease, thereby reducing the risk of heart disease, another cardiac event or death.
 - Cardiac rehabilitation programs include:
 - Counseling so the patient can understand and manage the disease process
 - Beginning an exercise program
 - Counseling on nutrition
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- Helping the patient modify risk factors such as high blood pressure, smoking, high blood cholesterol, physical inactivity, obesity and diabetes.
 - Providing vocational guidance to enable the patient to return to work
 - Supplying information on physical limitations
 - Lending emotional support
 - Counseling on appropriate use of prescribed medications



Post test

Mention
complication
post cardiac
surgery

Mention Nursing
care after open
heart surgery



Any question



Thank you